

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_modular_operator_diversity_residual_replay.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_034934_k
- started_at_local: 2026-05-20 03:49:34
- finished_at_local: 2026-05-20 04:23:26
- duration_hhmm: 00:34
- duration_seconds: 2031.168
- seed_offset: 10000
- replicate_label: k
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.903377	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235667	0.054968	0.121542	0.836758	0.4475	False	0.004357	0.330053
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235667	0.054968	0.121542	0.802005	0.4425	False	0.006979	0.336717
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235058	0.049643	0.117954	0.7365	0.4425	False	0.00524	1.51959

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.250653	0.068862	0.135838	0.0	0.4425	False	0.037997	0.682188
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235058	0.054968	0.121317	0.854962	0.4375	False	0.003506	0.003318

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.903377, complexity 0.56, and adaptation efficiency -0.027569.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.836758, complexity 0.4475, and adaptation efficiency 0.009715.
- Last-epoch runtime 42.410475 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.004357, positive scaffolds 1, Pareto runtime inflation 0.330053.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.235437, "gap_delta": 0.000224, "runtime_inflation": 0.330053, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.802005, complexity 0.4425, and adaptation efficiency 0.007279.
- Last-epoch runtime 73.323625 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta 0.006979, positive scaffolds 1, Pareto runtime inflation 0.336717.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.265747, "gap_delta": 0.000224, "runtime_inflation": 0.336717, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.049643, combined 0.117954.
- Accepted novelty 0.7365, complexity 0.4425, and adaptation efficiency 0.010363.
- Last-epoch runtime 50.252975 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.00524, positive scaffolds 2, Pareto runtime inflation 1.51959.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 1.664774, "gap_delta": -0.005462, "runtime_inflation": 1.51959, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.250653, family holdout 0.068862, combined 0.135838.
- Accepted novelty 0.0, complexity 0.4425, and adaptation efficiency 0.0.
- Last-epoch runtime 83.168375 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.037997, positive scaffolds 1, Pareto runtime inflation 0.682188.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 1.485397, "gap_delta": 0.017888, "runtime_inflation": 0.682188, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.854962, complexity 0.4375, and adaptation efficiency 0.01392.
- Last-epoch runtime 44.54995 ms and distance evaluations 4507.25.
- Validation: surviving False, transplant delta 0.003506, positive scaffolds 0, Pareto runtime inflation 0.003318.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.003318, "same_gap_faster": false}

Judge Appendix

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TSP Modular-Operator Discovery Suite: Conservative Interpretation

Summary

No discovered modular operators survive the critical transplant and ablation validations necessary to claim reusable operator structure or true transfer improvement.

Key Transfer and Hold-Out Gaps (Primary Transfer Evidence)

Condition	Final Family Gap	Final Transfer Gap	Final TSPLIB Gap	Transplant Mean Gap Delta	Surviving Candidate?
phase7_baseline_heuristic_only	0.05497	0.12132	0.23506	0.0	No
phase7_full_solver_evolution	0.06492	0.15940	0.21339	0.0	No
phase7_modular_operator_evolution	0.05497	0.12154	0.23567	0.00436	No
phase7_modular_operator_random_replay	0.05497	0.12154	0.23567	0.00698	No
phase7_modular_operator_diversity_residual_replay (best transfer condition)	**0.04964*	**0.11795**	**0.23506*	0.00524	No
phase7_modular_operator_compression_pressure	0.06886	0.13584	0.25065	0.03800	No
phase7_modular_operator_pareto_selection	0.05497	0.12132	0.23506	0.00351	No

Surviving Candidates

- ****None**** of the conditions report surviving candidates.

Modular Operators with Validation and Ablation Insights

1. Candidate Pruner Operators ("phase7_modular_operator_evolution" and "random_replay")

- ****Core Idea****: Adaptive candidate pruning tuned for mixed Euclidean 2-opt neighborhoods.
- ****Held-Out TSPLIB and Transfer Gaps****: No improvement vs. baseline heuristic.
- ****Transplant****: Minor positive gap delta (>0.004), but no clear improvement.
- ****Ablation****: Disabling operator causes no gap degradation, ablation does not support real operator effect.
- ****Runtime Inflation****: High ($\sim 33\%$ increase).
- ****Conclusion****: Validation fails transplant and ablation checks; no reusable operator confirmed.

2. Restart Controller Operator ("phase7_modular_operator_diversity_residual_replay")

- ****Core Idea****: Deterministic restart control adapting to stagnation with perturbation restarts.
- ****Held-Out Gaps****: Slightly better family gap (0.0496 vs 0.05497 baseline), but no TSPLIB improvement.
- ****Family Holdout Signal****: Improvement on "clustered_tsp" and "two_cluster_bottleneck_tsp" with -0.0046 mean gap delta.
- ****Transplant****: Positive gap delta (~ 0.0052), no improvement over baseline, fails clean transplant into some scaffolds.

- **Ablation**: Disabling operator worsens gap (+0.0055 gap delta) indicating some operator effect but no survival.
- **Runtime Inflation**: Very high (~152% increase).
- **Conclusion**: Shows some structured, reusable control idea, but does not survive transplant ablation fully or TSPLIB improvement; not a surviving modular operator.

3. Scaffold Selector Operator ("phase7_modular_operator_compression_pressure")

- **Core Idea**: Instance descriptor-based selection among baseline scaffolds.
- **Held-Out Gaps**: Worsens heldout TSPLIB gap substantially (gap delta +0.0198).
- **Transplant**: Large positive gap delta (~0.038), severe regressions in multiple families.
- **Ablation**: Disabling improves gap by ~0.0182, so operator likely harmful.
- **Runtime Inflation**: High (~68% increase).
- **Conclusion**: Fails all transfer and validation checks; not interesting.

4. Perturbation Operator ("phase7_modular_operator_pareto_selection")

- **Core Idea**: Deterministic stagnation escape with structured perturbations.
- **Held-Out Gaps**: No improvement over baseline.
- **Transplant**: Slight positive gap delta (~0.0035), no family improvement.
- **Ablation**: No effect when disabled (no gap delta).
- **Runtime Inflation**: Negligible (0.3%).
- **Conclusion**: No validated operator effect; fails survivor criteria.

Pareto Tradeoffs Analysis

- None of the modular operators show better gap at same or lower runtime than baseline.
- Most have substantial runtime inflation (especially restart controller and candidate pruner).
- No Pareto improvements justify operator deployment.

Conservative Conclusions

- **Held-Out TSPLIB Gap and Family Holdout Gap**: No modular operator improves these metrics conclusively.
- **Transplant and Ablation**: Operators fail to survive these critical checks; disabling operators often yields no degradation or even improvements.
- **Reusable Structure Evidence**: Weak or negative; no reusable operator structure established.
- **Full-Solver Performance**: The best full solver (phase7_full_solver_evolution) does not produce surviving modular operators.
- **Discovery Claim**: No modular operator survives scaffold/family transplant and ablation tests sufficiently to claim operator discovery or reusable module.

- ****Recommendation****: Current operators may inspire future designs but are not ready for impact or claim as discovered building blocks.

Summary Table: Operator Validation Status

Operator Name	Operator Type	Final Transfer Gap	Transplant Mean Gap Delta	Ablation Supported?	Runtime Inflation	Surviving Candidate?	Comments
candidate_pruner_transfer_focus_mixed_euc_grid_bottleneck	candidate_pruner	0.12154	0.00436	No	High (33%)	No	No clear benefit, fails ablation
instance_aware_candidate_pruner_transfer_stability_v1	candidate_pruner	0.12154	0.00698	No	High (34%)	No	Same as above
restart_controller_two_phase_bottleneck_v1	restart_controller	0.11795	0.00524	Partial (+0.00546 gap when disabled)	Very High (152%)	No	Some operator effect but costly
scaffold_selector_transfer_focus	scaffold_selector	0.13584	0.03800	No (disabling improves gap)	High (68%)	No	Harmful, fails transplant & ablation
transfer_bottleneck_escape_perturbation	perturbation	0.12132	0.00351	No	Negligible (0.3%)	No	No operator effect

Final Assessment

- The best transfer condition ("phase7_modular_operator_diversity_residual_replay") shows some operator effect but fails survival criteria for transplant and ablation and has too high runtime inflation.
- No operator qualifies as a robust, reusable discovered module.
- Full solver evolution yields best TSPLIB gap but no modular operator discovery.
- Operators are thus ****not yet validated for reuse or transfer claims****.